Tutorial - 3

10/02/2026

(Linear Regression)

Car Age (x_1) (m_1)	Mileage (x_2) (m_2)	Resale Price (Y)
2	3	6
3	4	7

$\Rightarrow y = mx + c$

$y = c + m_1 x_1 + m_2 x_2$

$\Downarrow$

$6 = c + m_1 \cdot 2 + m_2 \cdot 3$

$7 = c + m_1 \cdot 3 + m_2 \cdot 4$

matrix form:-

$$\underset{Y}{\begin{pmatrix} 6 \\ 7 \end{pmatrix}_{2\times 1}} = \underset{X}{\begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 4 \end{pmatrix}_{2\times 3}} \underset{\beta}{\begin{pmatrix} c \\ m_1 \\ m_2 \end{pmatrix}_{3\times 1}}$$

$$\boxed{Y = X\beta}$$

$\Rightarrow$ multiply X^T to the both sides

$X^T Y = (X^T X)\beta$ $\quad \because \left[(X^T X)^{-1} (X^T X) = 1 \right]$

$(X^T X)^{-1} (X^T Y) = \beta$

$$X^T = \begin{pmatrix} 1 & 1 \\ 2 & 3 \\ 3 & 4 \end{pmatrix}$$

$$X^T X = \begin{pmatrix} 1 & 1 \\ 2 & 3 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 4 \end{pmatrix}$$

$$= \begin{pmatrix} 1\times1 + 1\times1 & 1\times2 + 1\times3 & 1\times3 + 1\times4 \\ 2\times1 + 3\times1 & 2\times2 + 3\times3 & 2\times3 + 3\times4 \\ 3\times1 + 5\times1 & 3\times2 + 5\times3 & 3\times3 + 4\times4 \end{pmatrix}$$

$$= \begin{pmatrix} 1+1 & 2+3 & 3+4 \\ 2+3 & 4+9 & 6+12 \\ 3+5 & 6+15 & 9+16 \end{pmatrix} = \begin{pmatrix} 2 & 5 & 7 \\ 5 & 13 & 18 \\ 8 & 21 & 25 \end{pmatrix}$$

$$X^T X = \begin{pmatrix} 2 & 5 & 7 \\ 5 & 13 & 18 \\ 8 & 21 & 25 \end{pmatrix} = \begin{pmatrix} 2 & 5 & 10 \\ 5 & 13 & 21 \\ 8 & 21 & 34 \end{pmatrix}$$

$$(X^T X)^{-1} = \frac{\text{adj}\,(X^T X)}{\det\,(X^T X)}$$

$$\det\,(X^T X) = 2\,(13\times 34 - 21\times 21) - 5\,(5\times 34 - 8\times 21) + 10\,(5\times 21 - 8\times 13)$$

$$= 2 - 10 + 10$$

$$\det(X^T X) = 2$$

$$\text{adj}\,(X^T X) = \begin{bmatrix} +\begin{vmatrix} 13 & 21 \\ 21 & 34 \end{vmatrix} & -\begin{vmatrix} 5 & 21 \\ 8 & 34 \end{vmatrix} & +\begin{vmatrix} 5 & 13 \\ 8 & 21 \end{vmatrix} \\ -\begin{vmatrix} 5 & 10 \\ 21 & 34 \end{vmatrix} & +\begin{vmatrix} 2 & 10 \\ 8 & 34 \end{vmatrix} & -\begin{vmatrix} 2 & 5 \\ 8 & 21 \end{vmatrix} \\ +\begin{vmatrix} 5 & 10 \\ 13 & 21 \end{vmatrix} & -\begin{vmatrix} 2 & 10 \\ 5 & 21 \end{vmatrix} & +\begin{vmatrix} 2 & 5 \\ 5 & 13 \end{vmatrix} \end{bmatrix}$$

$$= \begin{bmatrix} (13\times 34 - 21\times 21) & (5\times 34 - 8\times 21) & (5\times 21 - 13\times 8) \\ (5\times 34 - 21\times 10) & (2\times 34 - 10\times 8) & (2\times 21 - 8\times 5) \\ (5\times 21 - 10\times 13) & (2\times 21 - 5\times 10) & (2\times 13 - 5\times 5) \end{bmatrix}$$

$$\begin{bmatrix} 1 & +2 & 1 \\ -40 & -12 & 2 \\ -25 & -8 & 1 \end{bmatrix}$$

$$(X^T X) = \frac{\text{adj}\,(X^T X)}{\text{det}\,(X^T X)}$$

$$X^T y = \underset{3\times 2}{\begin{bmatrix} 1 & 1 \\ 2 & 3 \\ 3 & 5 \end{bmatrix}} \underset{2\times 1}{\begin{bmatrix} 6 \\ 7 \end{bmatrix}} = \begin{bmatrix} 6+7 \\ 12+21 \\ 18+35 \end{bmatrix} = \begin{bmatrix} 13 \\ 33 \\ 53 \end{bmatrix}$$

$$= \underset{3\times 3}{\begin{bmatrix} 0.5 & 20 & -12.5 \\ -1 & -6 & 4 \\ 0.5 & -1 & 0.5 \end{bmatrix}} \begin{bmatrix} 13 \\ 33 \\ 53 \end{bmatrix}$$

$$= \begin{bmatrix} 0.5\times 13 + 20\times 33 + (-12.5\times 53) \\ -1\times 13 + -6\times 33 + 4\times 53 \\ 0.5\times 13 + -1\times 33 + 0.5\times 53 \end{bmatrix}$$

$$= \begin{bmatrix} 4 \\ 1 \\ 0 \end{bmatrix} \Rightarrow \beta$$

$c = 4$

$m_1 = 1$

$m_2 = 0$

$$\hat{y} = c + m_1 x_1 + m_2 x_2$$

$$\boxed{\hat{y}_{pred} = 4 + 1 \cdot x_1 + 0 x_2}$$

$\hat{y}_1 = 4 + 2\cdot(1) + 3(0)$

$$\boxed{\hat{y}_{1\,pred} = 6.}$$

$\hat{y}_2 = 4 + (3)1 + 0\cdot(5)$ } as these values are equal to actual values so the a. is '0'.

$= 7$

$$\boxed{\hat{y}_{2\,pre} = 7}$$

Now calculate the a. value

→ Root mean square Error =-

$\Rightarrow \frac{1}{N} \sum_i (y_i - \hat{y}_i)^2$

$= \frac{1}{2} \sum_i [(0)^2 + (0)^2]$

$= \frac{1}{2} [(0)^2 + (0)^2]$

$= 0$

Ex: $\hat{y} = 2$

$\hat{y} = 7$

then.

$= \frac{1}{2} [(6-2)^2 + (7-7)^2]$

$= \frac{1}{2} [(4)^2 + (0)^2]$

$= \frac{1}{2} \cancel{168}$

$= 8$

6 - 6

6 - 6

① which variable act as predictors?

⇒ x_1 is important as x_2 coefficient is '0' so do not importance of x_2.

②